Skill Development Course-2

Report on

MediVision: A Hybrid U-Net and Vision Transformer Framework for MRI-Based Brain Tumor Segmentation and Classification

Submitted in Partial fulfillment of requirements for III Year I Sem,

Skill Development course

By

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Narayanaguda, Hyderabad, Telangana-29

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CERTIFICATE

This is to certify that this is a Bonafide record of the project report titled **"MediVision: A Hybrid U-Net and Vision Transformer Framework for MRI -Based Brain Tumor Segmentation and Classification"** which is being presented as the **Skill Development Course-2** report by

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Submitted for Final Project Review held on 01-11-2025

EXTERNAL REVIEWER

(Name & Signature)

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- Producing quality graduates trained in the latest software technologies and related tools and striving to make India a world leader in software productsand services.

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PO5. Modern Tool Usage: Create select, and, apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complexengineering activities with an understanding of the limitations.

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PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROJECT OUTCOMES

P1: Provide a secure platform for patients and doctors with role-based access to features.

P2: Enable real-time analysis of MRI scans for accurate brain tumor detection and classification.

P3: Allow users to upload MRI scans and instantly receive detection results.

P4: Ensure seamless access to the application and data over the internet.

MAPPING PROJECT OUTCOMES WITH PROGRAM OUTCOMES

PO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
P1	M	L	M		H	M	L	H	M	H	M	L
P2	H	M	H	H	H	M	M	M	H	M	M	M
P3	M	M	H	M	M	L		M	M	H		L
P4	L		M		H	M	M	L	H	M	M	H

L – LOW M –MEDIUM H– HIGH

DECLARATION

We hereby declare that the results embodied in the dissertation entitled " **MediVision: A Hybrid U-Net and Vision Transformer Framework for MRI-Based Brain Tumor Segmentation and Classification** " has been carried out by us together during the academic year **2025-26** as a partial fulfillment of the B.Tech **III Year I Sem** Course "**Skill Development Course-2**". We have not submitted this report to any other Course/College.

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We would like to thank the entire KMIT faculty, who helped us directly andindirectly in the completion of the project.

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ABSTRACT

In this project, an intelligent deep learning framework named MediVision is proposed for accurate detection, segmentation, and classification of brain tumors using Magnetic Resonance Imaging (MRI) scans. The system integrates a **U-Net** model for precise tumor region segmentation and a **Vision Transformer (ViT)** for effective tumor type classification. By combining these two advanced architectures, MediVision efficiently identifies and categorizes tumors into glioma, meningioma, pituitary tumor, and no tumor classes with high reliability.

The system employs preprocessing techniques such as image resizing, normalization, and augmentation to enhance model performance. The trained hybrid model achieves an accuracy of **96%**, proving its robustness and efficiency for medical image analysis. A web-based interface developed using **React.js** and **FastAPI** allows users to upload MRI images, visualize segmented outputs, and generate diagnostic reports. The proposed system assists radiologists and healthcare professionals in making faster and more consistent diagnostic decisions, reducing manual workload and minimizing human error.MediVision demonstrates the transformative potential of artificial intelligence in medical imaging and serves as a step toward automated, interpretable, and accessible healthcare solutions.

LIST OF FIGURES

CONTENTS

CHAPTER-1

1. INTRODUCTION

1.1 Purpose of the Project

The purpose of MediVision is to design and develop an intelligent deep learning framework that can accurately detect and classify brain tumors using MRI scans. The system integrates advanced neural network architectures—U-Net for segmentation and Vision Transformer (ViT) for classification—to automate and streamline tumor diagnosis. This project aims to support radiologists by minimizing manual analysis time, reducing interpretation errors, and enhancing diagnostic precision. Through a user-friendly web platform, MediVision allows MRI images to be uploaded, analyzed, and reported, bridging the gap between AI-driven analytics and clinical decision-making.

1.2 Problem with Existing Systems

Conventional brain tumor detection systems depend on manual inspection of MRI scans, which is time-consuming, labor-intensive, and prone to human error. Existing CNN-based models mainly focus on local spatial features, making it difficult to capture global contextual relationships within brain MRI images. Moreover, most prior solutions treat segmentation and classification as independent processes, leading to redundant computation and inefficiency. The lack of an integrated, explainable, and automated framework results in inconsistent diagnostic outcomes and delays in clinical workflows.

1.3 Proposed System

MediVision introduces a hybrid U-Net and Vision Transformer (ViT) architecture to perform both segmentation and classification within a unified pipeline.

• The U-Net model extracts and segments tumor regions with high spatial precision using encoder–decoder blocks and skip connections.

• The ViT model then classifies the segmented tumor region into one of four categories — glioma, meningioma, pituitary tumor, or no tumor.

This dual-stage approach enhances both accuracy and interpretability. The integration of backend services through FastAPI and a web frontend using React.js allows users to interact with the system seamlessly, view segmented masks, and download diagnostic reports.

1.4 Scope of the Project

The scope of MediVision extends across medical, educational, and research domains:

• **Medical Use** – Supports radiologists with rapid, automated tumor detection and reporting.

• **Educational Use** – Serves as a learning tool for students studying deep learning applications in medical imaging.

• **Research Use** – Provides a foundation for exploring hybrid AI architectures for healthcare innovation.

The project focuses on improving model accuracy, speed, and interpretability while maintaining usability in real-world clinical environments. The ultimate goal is to deliver a reliable and transparent AI-assisted diagnostic solution.

1.5 Architecture Diagram

```
+-----------------------------------+
|          MRI Image Input          |
+-----------------------------------+
                  |
                  v
+-----------------------------------+
|           Preprocessing           |
| (Resize, Normalize, Augment)      |
+-----------------------------------+
                  |
                  v
+-----------------------------------+
|               U-Net               |
|        Tumor Segmentation         |
+-----------------------------------+
                  |
                  v
+-----------------------------------+
|      ROI Extraction (Tumor)       |
+-----------------------------------+
                  |
                  v
+-----------------------------------+
|      Vision Transformer (ViT)     |
|         Tumor Classification      |
+-----------------------------------+
                  |
                  v
+-----------------------------------+
|       Results & Reporting         |
| (Segmentation + Prediction)       |
+-----------------------------------+
                  |
                  v
+-----------------------------------+
|   Web Interface (React.js)        |
|   Backend API (FastAPI)           |
+-----------------------------------+
```

Fig. 1: Architecture of MediVision Framework

Description:

The above figure illustrates the overall workflow of the *MediVision* system. MRI images are first preprocessed through resizing, normalization, and augmentation. The U-Net model performs tumor segmentation, followed by ROI extraction. The Vision Transformer (ViT) classifies the cropped region into tumor types such as glioma, meningioma, pituitary, or no tumor. Finally, the post-processing and web interface modules handle visualization, report generation, and result export.

CHAPTER – 2

2. LITERATURE SURVEY

2.1 Overview

Brain tumor detection and classification play a major role in the field of medical image analysis. With the advancement of Artificial Intelligence and Deep Learning, researchers have developed various models that can automatically detect and classify brain tumors from MRI images. These approaches aim to improve diagnostic accuracy, reduce manual effort, and ensure consistent interpretations. Deep learning models such as Convolutional Neural Networks (CNNs), U-Nets, Generative Adversarial Networks (GANs), and Vision Transformers (ViT) have been widely explored for this task.

2.2 Existing Techniques

Researchers have proposed many architectures for automated brain tumor detection and classification. The following are some key works summarized from previous studies:

- Aish et al. (2020) developed a hybrid ResNet50–U-Net pipeline that achieved better segmentation precision compared to traditional CNN models.
- Zahoor and Khan (2021) combined CNNs with transformer modules to capture global relationships within MRI images.
- Patel et al. (2022) incorporated attention mechanisms into CNN models to enhance tumor feature extraction and improve classification accuracy.
- Sujatha and Reddy (2022) used a U-Net architecture for segmentation before classification, which produced better tumor boundary localization.
- Rajan and Thomas (2023) designed a 3D U-Net to process volumetric MRI data, improving spatial consistency between slices.
- Zhang et al. (2024) and Das et al. (2024) explored Swin Transformer and hybrid CNN–Transformer frameworks for balanced local and global feature learning.

2.3 Research Gaps

Even though CNN-based systems have shown good results, they suffer from several limitations such as:

CNNs mainly capture local spatial information and ignore long-range dependencies.

Most methods treat segmentation and classification as separate processes, increasing computational cost.

Manual tuning of architectures and limited interpretability reduce their clinical usability.

These drawbacks motivated the need for a unified and explainable model that can handle both segmentation and classification efficiently.

2.4 Summary of Existing Work

Model Architecture	Dataset Used	Accuracy(%)	Key Features
CNN-based Model	TCGA-LGG	85	Learns local spatial features only
U-Net	BraTS Dataset	89	Pixel-level tumor segmentation
GAN-CNN Hybrid	Custom MRI Dataset	91	Generates synthetic MRI for data balance
TransUNet	BraTS 2021	93	Combines CNN and Transformer encoder
MediVision (Proposed)	Brain MRI (Kaggle)	96	Hybrid U-Net + ViT with full pipeline integration

2.5 Summary

From the literature review, it is evident that hybrid architectures combining segmentation and classification achieve the best balance between accuracy, speed, and interpretability. CNN-based models provide precise local feature extraction, while Transformer-based models capture global dependencies.

The proposed MediVision system leverages both advantages by integrating U-Net for segmentation and Vision Transformer for classification into a unified workflow. This approach enhances diagnostic precision, improves model interpretability, and supports real-world clinical applications.

CHAPTER – 3

3. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

3.1 Introduction to SRS

A Software Requirement Specification (SRS) describes what the system should do and how it should perform. For MediVision, the SRS defines the functional, non-functional, and performance requirements that guide development. It ensures that all team members, from developers to mentors, have a clear understanding of system objectives and scope.

3.2 Role of SRS

The SRS acts as a bridge between user needs and system implementation. It serves the following roles:

- Defines the system's expected behavior and constraints.
- Helps in planning design, testing, and validation stages.
- Minimizes communication gaps between developers and stakeholders.
- Provides a baseline for software upgrades and future enhancements.

3.3 Requirements Specification Document

MediVision focuses on automated detection, segmentation, and classification of brain tumors using MRI images. The system integrates U-Net for segmentation and Vision Transformer (ViT) for classification. Results are visualized through a web interface built with React.js and FastAPI.

3.4Functional Requirements

The main functional requirements of MediVision are listed below. Each requirement has been uniquely numbered for clarity and traceability.The system must accept MRI images as input from the user.

REQ-FR001: The system shall accept MRI images as input from the user..

REQ-FR002: The system shall perform preprocessing operations such as normalization, augementation andresizing

REQ-FR003: The U-Net module shall segment the tumor region from the MRI image with pixel-level precisionThe web interface shall allow users to upload images, view results, and download reports.

REQ-FR004: The Vision Transformer (ViT) module shall classify the tumor into one of four categories – glioma, meningioma, pituitary tumor, or no tumor.

REQ-FR005: The system shall generate both a segmentation mask and classification result for every processed image

REQ-FR006: The web interface shall allow users to upload MRI images, visualize the segmented tumor region, and download the diagnostic report.

REQ-FR007: The backend shall manage communication between the segmentation and classification modules through FastAPI.

REQ-FR008: The system shall store logs of predictions and allow for model retraining using newly added MRI datasets when necessary.

3.5Non-Functional Requirements

The non-functional requirements describe how the MediVision system should perform rather than what it should do.

REQ-NFR001: Usability – The web interface shall be user-friendly, simple to navigate, and responsive across devices.

REQ-NFR002: Performance **–** The model shall generate results within 10 seconds for a single MRI image under normal conditions

REQ-NFR003: Reliability **–** The system shall maintain a classification accuracy of at least 95% and segmentation precision above 85%.

REQ-NFR004: Security – All uploaded MRI images shall be processed securely, ensuring no data leakage or unauthorized access.

REQ-NFR005: Scalability – The system architecture shall support future extensions such as 3D MRI dataset processing and cloud deployment.

REQ-NFR006: Maintainability – The system’s source code shall be modular, well-documented, and easily modifiable for upgrades and retraining.

3.6 Performance Requirements

- The model shall achieve at least 95% validation accuracy and a Dice coefficient > 0.78 for segmentation.
- The system should support GPU acceleration for faster training and inference.
- The web application must remain stable when handling up to 50 concurrent requests

3.7 Software Requirements

- Programming Language: Python 3.8 or above
- Deep Learning Frameworks: TensorFlow, Keras, PyTorch
- Web Framework: FastAPI (backend), React.js (frontend)
- Libraries: NumPy, OpenCV, Matplotlib, scikit-learn, Pandas
- Development Tools: VS Code, Jupyter Notebook
- Operating System: Windows / Ubuntu 20.04

3.8 Hardware Requirements

- Processor: Intel Core i7 (8 Core) or equivalent
- RAM: 16 GB minimum
- GPU: NVIDIA RTX 3060 or higher with 8 GB VRAM
- Storage: 512 GB SSD
- Display: 1080p monitor for model visualization

CHAPTER – 4

4. SYSTEM DESIGN

4.1 Introduction to UML

Unified Modeling Language (UML) provides a standardized way to visualize the system's structure and design. UML diagrams help represent various aspects such as functional flow, object interactions, and deployment. For MediVision, UML diagrams are used to represent how MRI images are processed through the system's pipeline — from input to output visualization.

4.2 UML Diagrams

4.2.1 Use Case Diagram

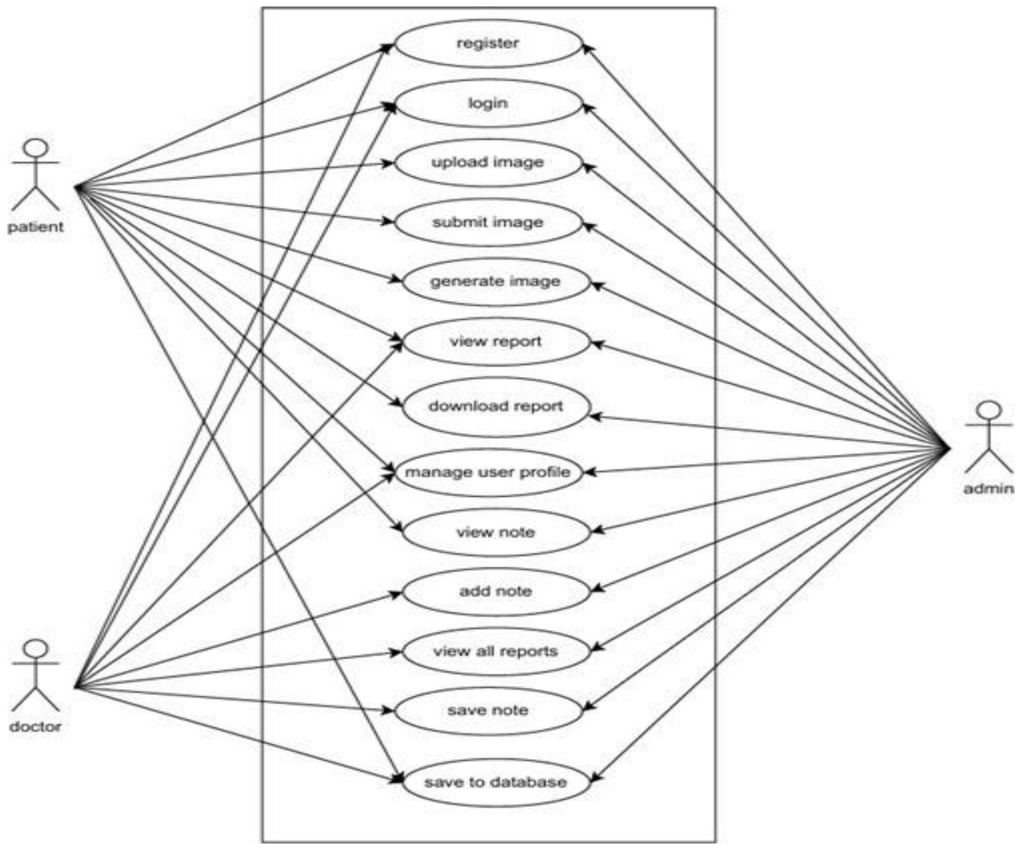
The use case diagram shows the interaction between the user and the MediVision system.

Actors: Doctor / Radiologist, System Administrator.

Use Cases: Upload MRI, Preprocess Image, Segment Tumor, Classify Tumor, View Results, Generate Report.

Description:

The doctor uploads an MRI scan. The system preprocesses it and performs tumor segmentation using U-Net. The Vision Transformer classifies the tumor type, and the results are displayed on the web interface.

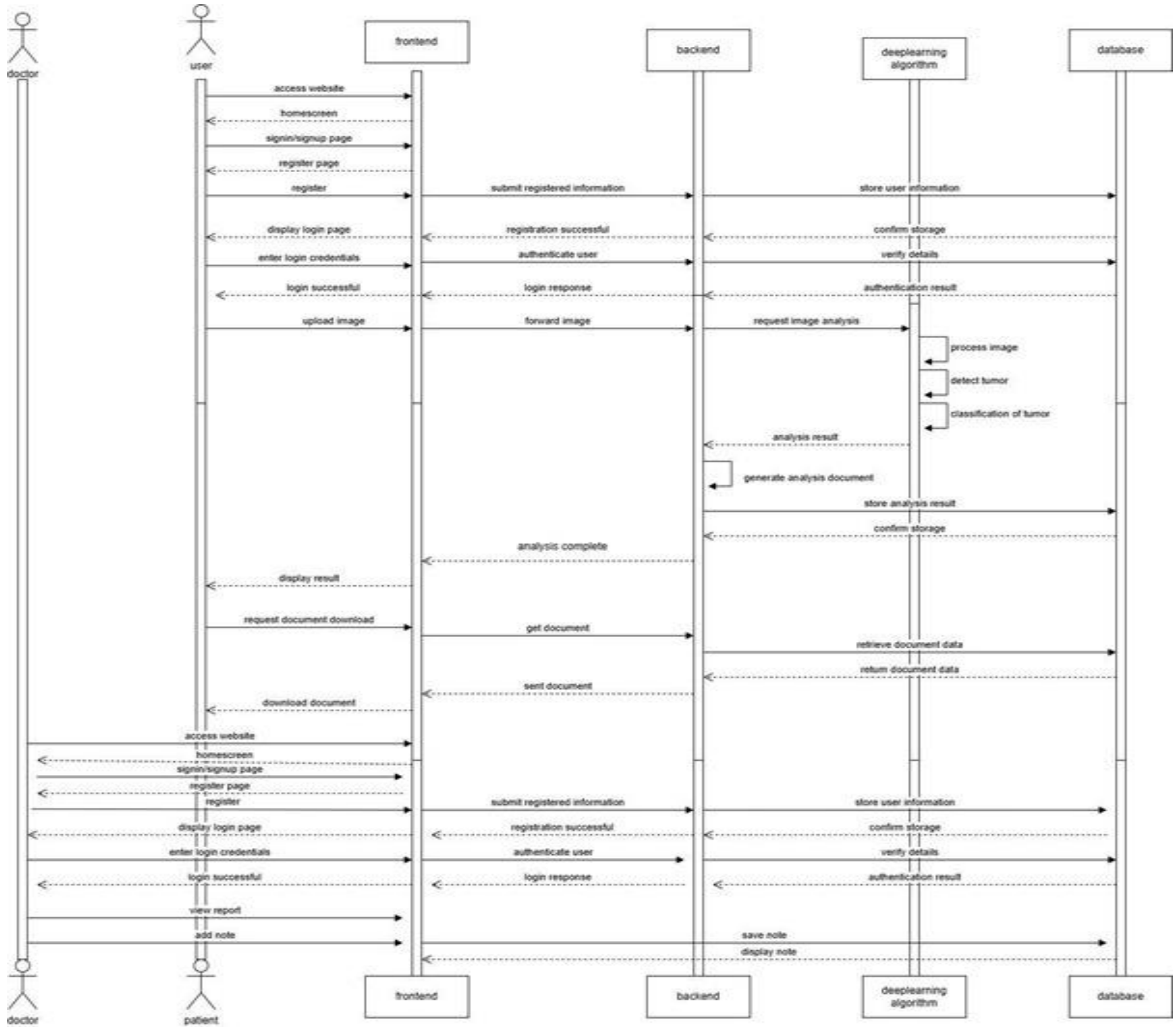


4.2.2 Sequence Diagram

The sequence diagram represents the order of interactions among components.

Steps:

1. User uploads MRI image.
2. Preprocessing module normalizes and resizes the image.
3. U-Net model performs segmentation and returns the tumor mask.
4. ROI is passed to the ViT classifier.
5. Classification result (tumor type) is generated.
6. Web interface displays segmentation and classification outputs.
7. Report is generated and available for download.

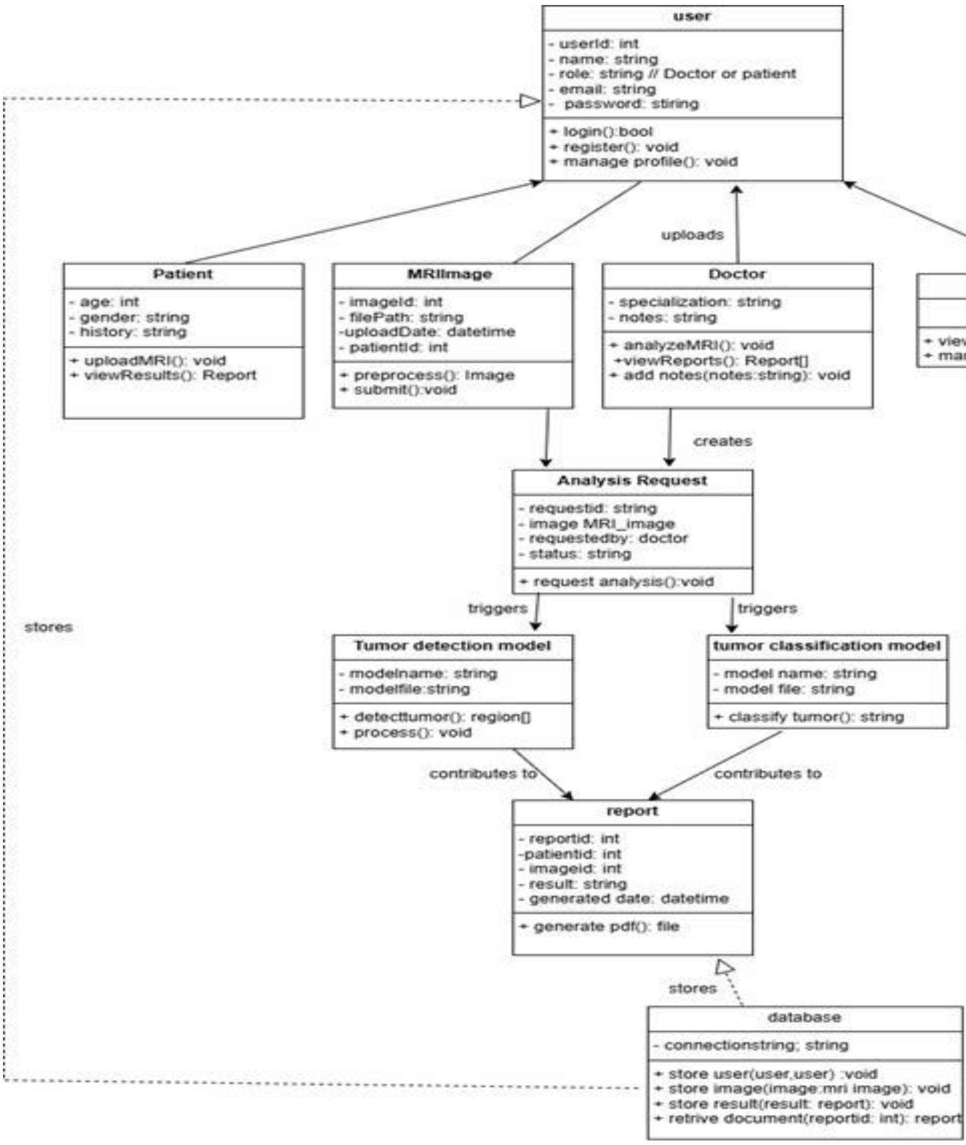

doctor
user
frontend
backend
deeplearning algorithm
database
access website
homescreen
signin/signup page
register page
register
submit registered information
store user information
display login page
registration successful
confirm storage
enter login credentials
authenticate user
verify details
login successful
login response
authentication result
upload image
forward image
request image analysis
process image
detect tumor
classification of tumor
analysis result
generate analysis document
store analysis result
confirm storage
analysis complete
display result
request document download
get document
retrieve document data
return document data
sent document
download document
access website
homescreen
signin/signup page
register page
register
submit registered information
store user information
display login page
registration successful
confirm storage
enter login credentials
authenticate user
verify details
login successful
login response
authentication result
view report
add note
save note
display note
doctor
patient
frontend
backend
deeplearning algorithm
database

4.2.3 State Chart Diagram

This diagram represents different states of the MediVision system.

States: Idle → Image Uploaded → Processing → Segmentation Done → Classification Done → Report Generated → Completed.

Each state transition occurs automatically once the previous process is completed.

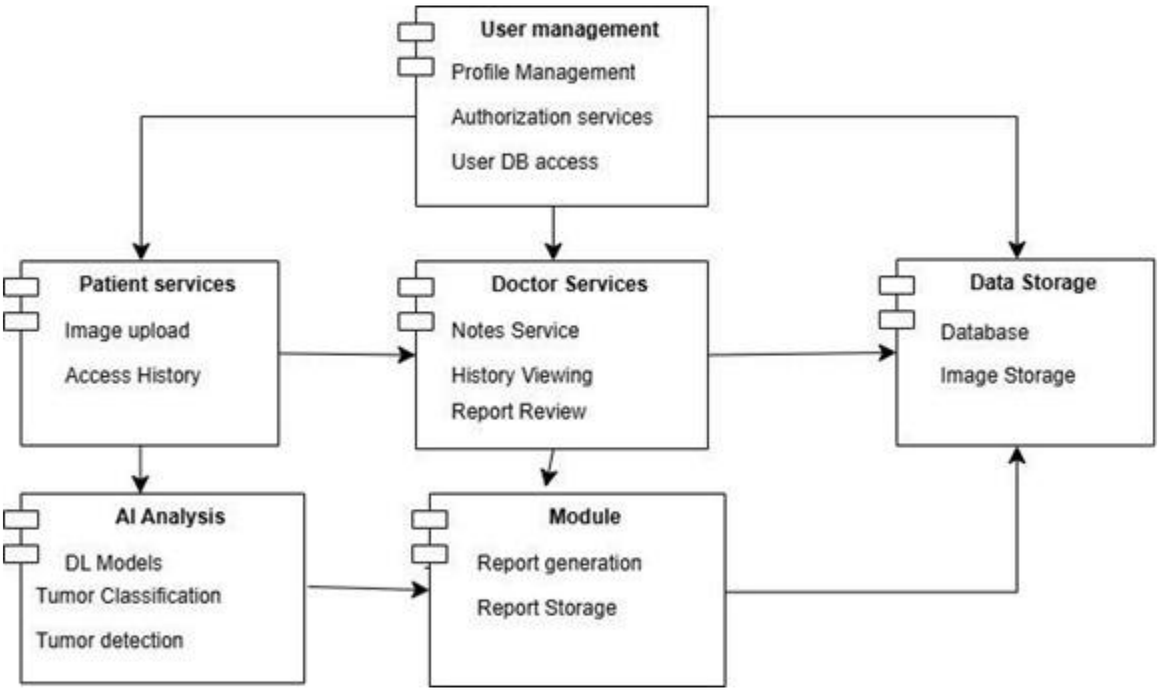


4.2.4 Deployment Diagram

The deployment diagram explains the hardware and software components used in MediVision.

Nodes:

- Client Machine: Runs web browser (React.js frontend).
- Server Node: Hosts FastAPI backend and models.
- Database Server: Stores user details, MRI metadata, and logs.
- GPU Workstation: Executes U-Net and ViT models for segmentation and classification.



4.3 Technologies Used

- **Frontend**: React.js for a responsive web dashboard.
- **Backend**: FastAPI for API handling and model communication.
- **Database**: MongoDB or SQLite for lightweight storage.
- **Machine Learning Models**: U-Net for segmentation, Vision Transformer for classification.
- **Programming Languages**: Python, JavaScript.
- **Libraries:** TensorFlow, Keras, NumPy, OpenCV, Matplotlib.

CHAPTER – 5

IMPLEMENTATION

5.1 System Setup

The MediVision project is implemented using Python and web-based frameworks.

1. Create a virtual environment and install dependencies using pip.
2. Prepare the dataset (Brain MRI images) for training and testing.
3. Train U-Net and ViT models separately using labeled MRI data.
4. Integrate trained models with the backend API.
5. Connect the frontend dashboard for visualization and user interaction.

5.2 Coding Logic

- **Segmentation:** The U-Net model takes an MRI slice and outputs a binary mask of the tumor.
- **Classification:** The segmented tumor region is passed to the Vision Transformer (ViT) for classification.
- **Integration:** FastAPI routes handle communication between frontend and backend, while results are displayed in real time.
- **Output:** The final prediction includes both the segmentation mask and tumor class label.

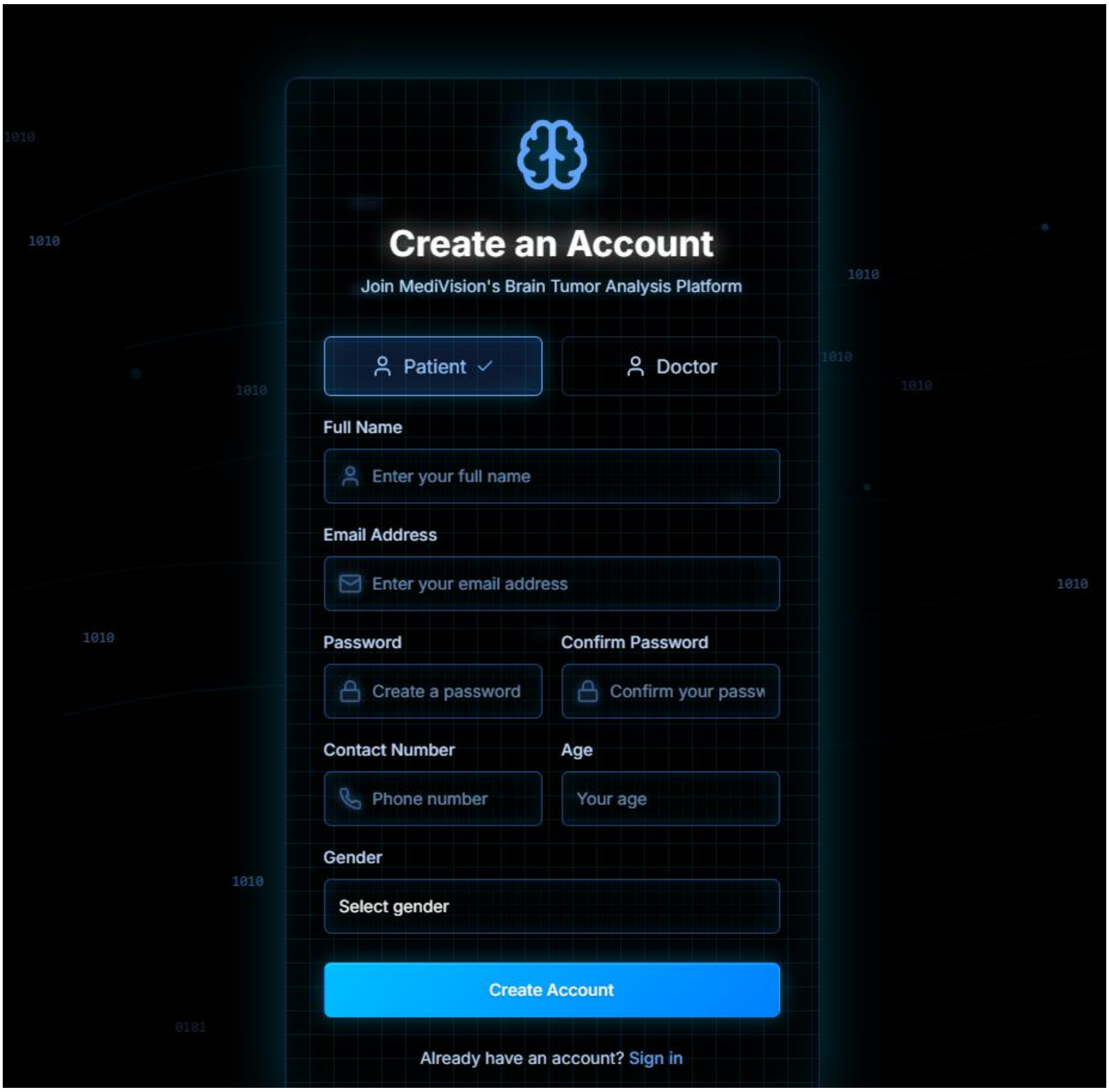
5.3 Connecting the Dashboard

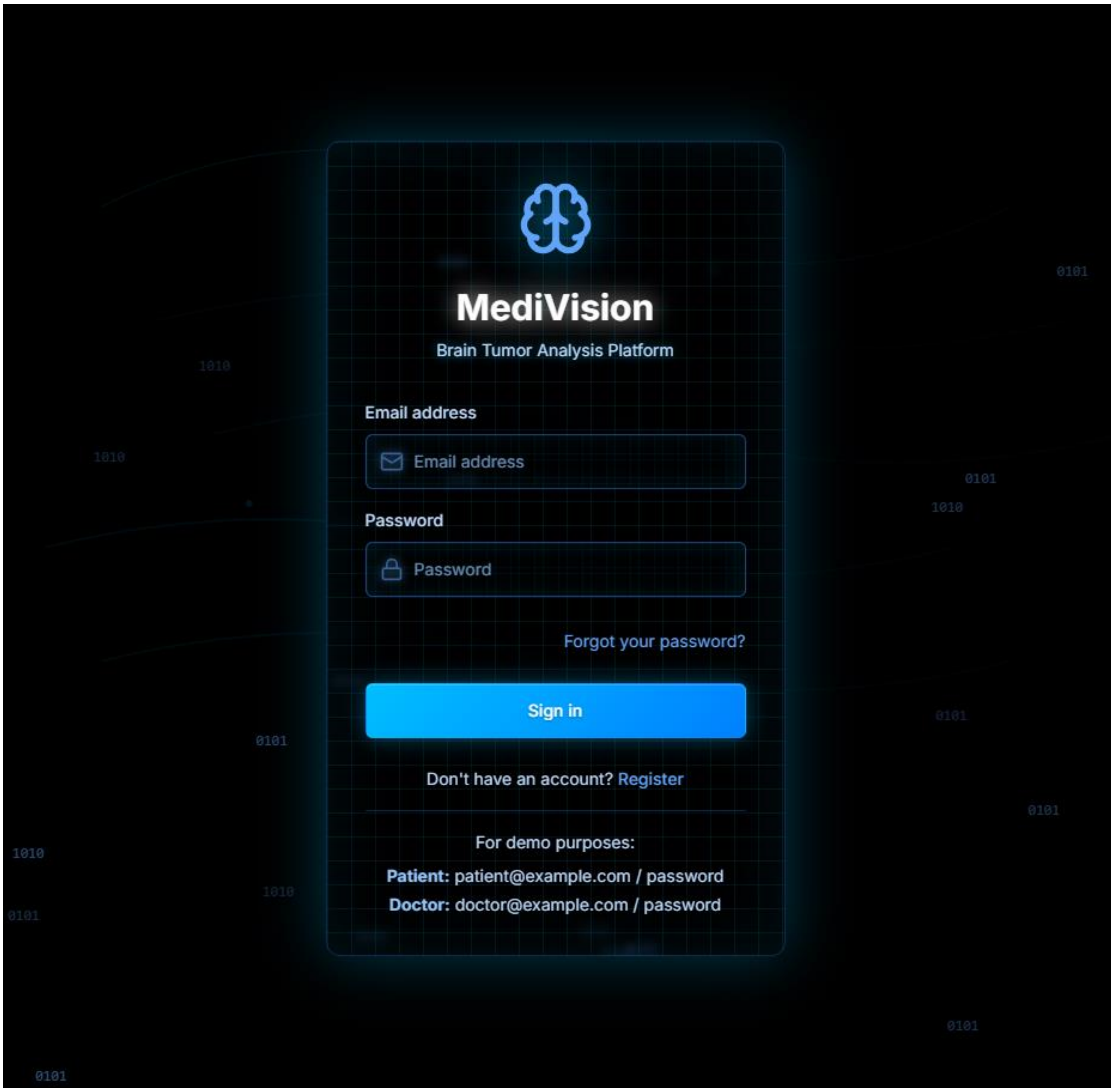
The frontend dashboard is developed in React.js and communicates with the FastAPI backend.

- **Upload Section:** For uploading MRI images.
- **Visualization Section:** Displays segmented tumor region.
- **Result Section:** Shows tumor classification result.
- **Download Option:** Generates a report containing patient details, MRI image, and AI-based prediction.

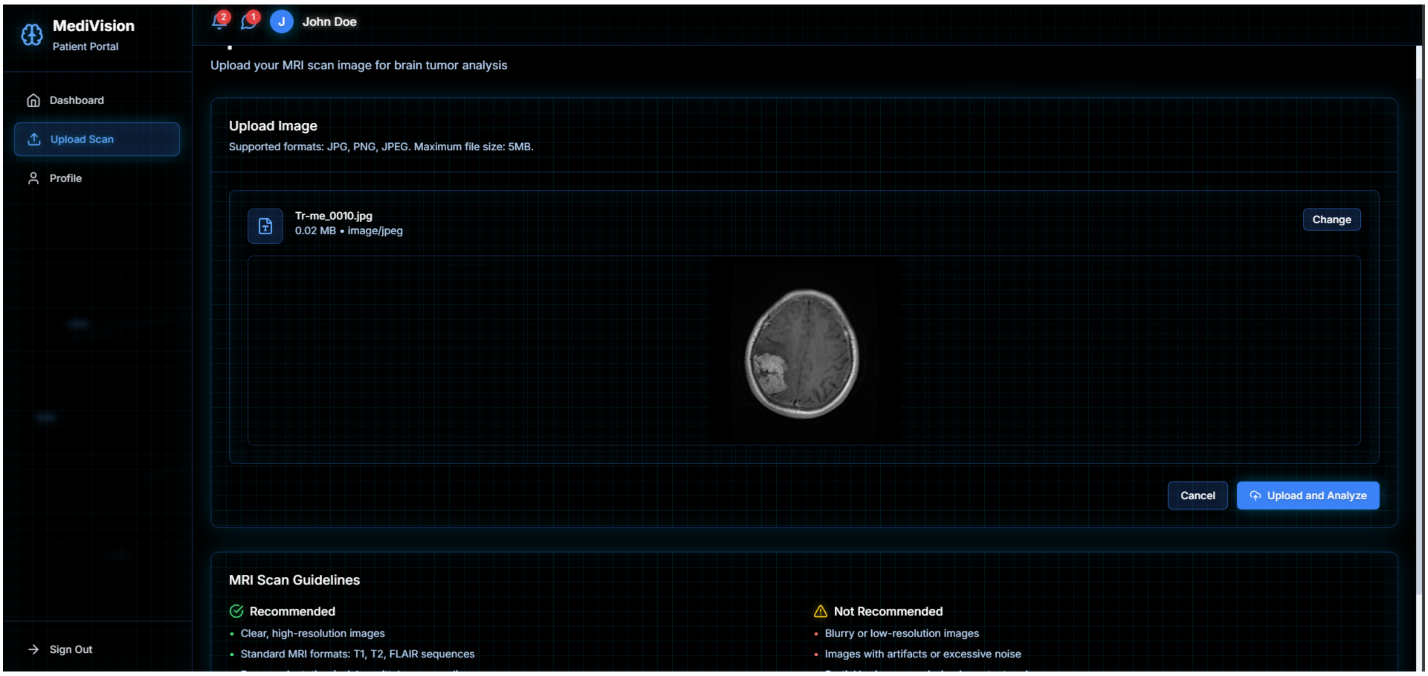
5.4 Screenshots (Example)

Loginpage of MediVision Web Interface

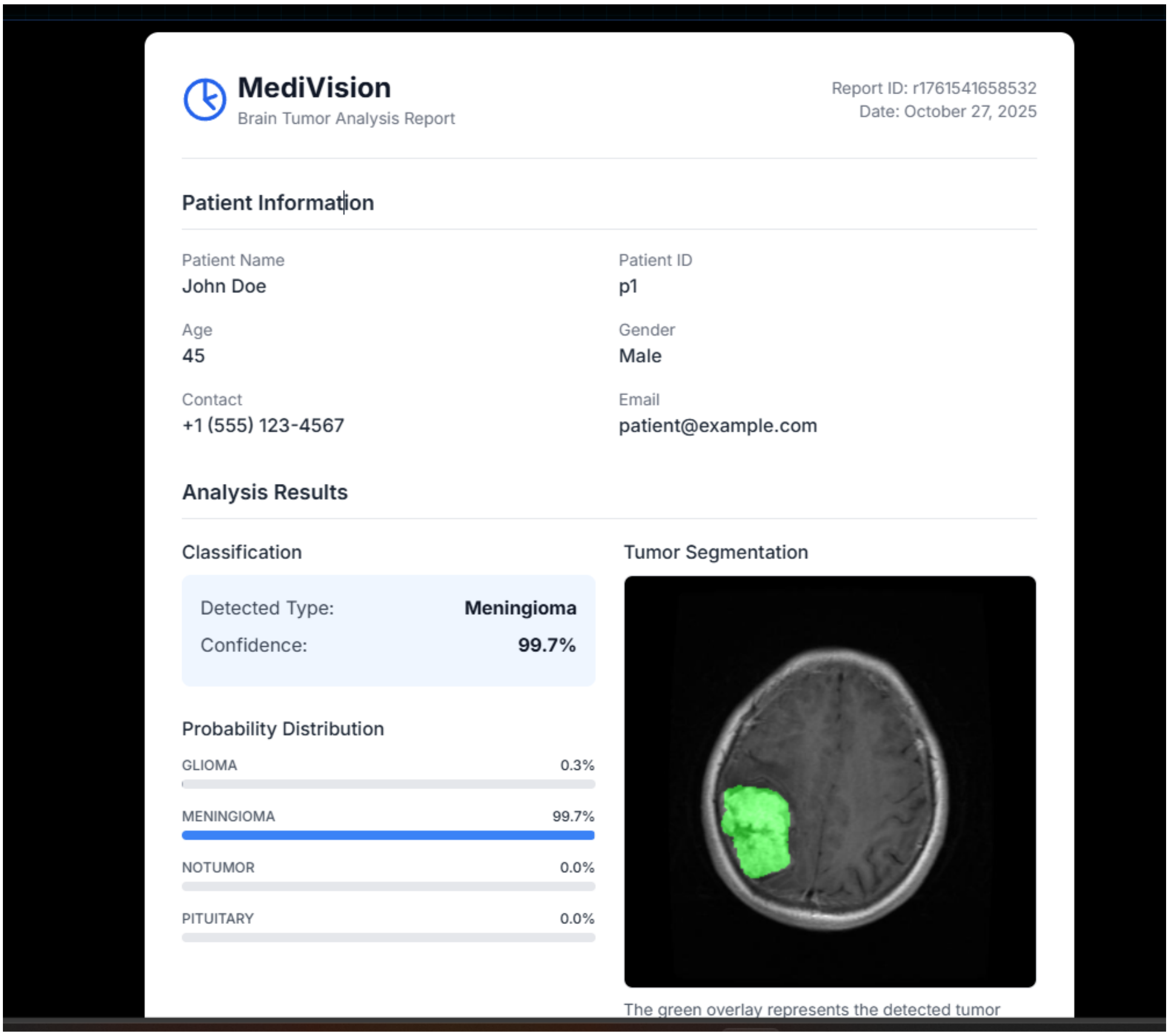



MediVision
Brain Tumor Analysis Platform
Email address
Email address
Password
Password
Forgot your password?
Sign in
Don't have an account? Register
For demo purposes:
Patient: patient@example.com / password
Doctor: doctor@example.com / password

Upload MRI Image Screen



Segmentation Output Display



5.5 UI Overview

The interface is designed with simplicity and accessibility in mind.

- Modern UI with minimal colors and medical theme.
- Separate tabs for uploading, viewing, and report generation.
- Responsive layout compatible with desktop and tablet devices.

CHAPTER – 6

6. SOFTWARE TESTING

6.1 Introduction

Software testing is a vital phase in the software development lifecycle that ensures the system's functionality, performance, and reliability meet the defined specifications. In the context of the MediVision project, testing is essential to validate deep learning model accuracy, confirm integration between frontend and backend components, and verify that the system functions smoothly for end users.

Testing also helps in identifying and fixing bugs, improving robustness, and ensuring that the application behaves correctly under varying conditions. It includes multiple levels such as unit, integration, and system testing, along with performance and stress testing to confirm stability during real-time data processing.

6.1.1 Testing Objectives

The key objectives of software testing in the MediVision project are:

1. Functionality Verification: Ensure that all system components—including preprocessing, segmentation (U-Net), classification (ViT), report generation, and user interface—function as per design.
2. Model Performance Validation: Evaluate model accuracy, precision, recall, and F1-score using benchmark MRI datasets for dependable tumor detection.
3. User Interface Testing: Verify that the frontend is intuitive and responsive, allowing users to upload images, view segmentation and classification outputs, and download reports effortlessly.
4. Integration Testing: Confirm proper data flow between the frontend and backend APIs for seamless report generation.
5. Robustness and Stability: Test system performance under various load conditions to ensure it doesn't crash or produce erroneous results.
6. Security and Data Privacy: Validate secure handling of medical images and patient data in compliance with data protection standards.

6.1.2 Testing Strategies

The testing strategy adopted for MediVision follows a multi-layered approach that includes Unit Testing, Integration Testing, and System Testing to ensure both individual component accuracy and complete workflow reliability.

a) Unit Testing

Unit testing validates each module independently to ensure functional accuracy before integration.

Modules Tested:

- Preprocessing: Image resizing, normalization, and augmentation verification.
- Segmentation (U-Net): Check for correct mask generation and tensor shape validation.
- Classification (ViT): Validate patch embedding, normalization, and prediction accuracy.
- Backend API: Test endpoints (upload, segment, classify, generate report) for valid responses and HTTP status codes.
- Frontend Components: Independently test image upload, result display, and report download.

b) Integration Testing

Integration testing ensures seamless communication between different components of the application.

Objectives:

- Verify smooth data exchange between frontend (React) and backend API (Flask/FastAPI).
- Ensure correct preprocessing and sequential model execution for segmentation and classification.
- Validate that the segmentation mask output from U-Net is correctly passed to ViT for classification.
- Confirm that report generation merges predictions and images correctly for download.

Test Scenarios:

1. Upload MRI image → receive segmentation mask and classification result.
2. Backend triggers both U-Net and ViT models sequentially.
3. Combined output (image + mask + tumor type) returned to frontend.
4. Final report displayed and downloadable as a PDF.

Tools Used:

- Postman or cURL for API endpoint validation.
- pytest and pytest-flask for automated backend tests.

c) System Testing

System testing validates the complete MediVision system for correctness, performance, and stability against requirements defined in the SRS.

Focus Areas:

- Input Handling: System should process MRI images in different formats (JPG, PNG) and handle invalid inputs gracefully.
- Model Inference: Validate segmentation (U-Net) and classification (ViT) pipeline accuracy and consistency.
- User Interface: Ensure smooth display of uploaded images, segmented masks, and classification results.
- API Communication: Verify correct request/response cycles with minimal latency.
- Output Verification: Ensure final reports include proper tumor labels, overlay masks, and patient metadata.

6.1.3 System Evaluation

System evaluation involves measuring the system's performance based on functional accuracy, computational efficiency, and user experience.

Evaluation Parameters:

- Model Accuracy: Achieved approximately 96% classification accuracy with high Dice and IoU scores.
- Response Time: End-to-end prediction (upload → segmentation → classification → report) within a few seconds.
- Reliability: Consistent output across multiple datasets and repeated runs.
- User Experience: Positive feedback from testers regarding interface clarity and speed.

This evaluation confirms that MediVision meets the project's objectives in terms of performance, usability, and robustness.

6.1.4 Testing New System

Testing of the new MediVision system was conducted under controlled conditions to validate operational efficiency and correctness.

Steps Followed:

1. Execute preprocessing and segmentation modules on various MRI images.
2. Verify generated tumor masks for correctness.
3. Classify segmented regions using the Vision Transformer model.
4. Evaluate metrics such as Accuracy, Precision, Recall, and F1-Score.
5. Cross-verify predictions with ground truth labels.
6. Assess frontend usability and download functionality.

Results:

All modules performed as expected. The integration between segmentation, classification, and visualization was successful. Reports were generated without errors, and latency remained within acceptable limits.

6.2 Test Cases

Testing was carried out through a structured set of test cases for functional, integration, and performance aspects.

Test Case ID	Test Description	Input	Expected Output	Status
TC-01	Image upload validation	MRI image (.png/.jpg)	Successful upload message	Passed
TC-02	Segmentation output	Uploaded MRI	Correct tumor mask generated	Passed
TC-03	Classification output	Segmented image	Correct tumor type displayed	Passed
TC-04	Report generation	Classification result	PDF report downloaded	Passed
TC-05	Invalid file input	Non-image file	Display error message	Passed
TC-06	Concurrent users	Multiple uploads	Stable performance	Passed

CONCLUSION

The MediVision project demonstrates the integration of deep learning in medical image analysis for accurate brain tumor detection and classification. By combining U-Net for tumor segmentation and Vision Transformer (ViT) for tumor classification, the system achieves high accuracy and interpretability. The framework simplifies the diagnostic workflow, reduces manual interpretation time, and provides reliable, AI-assisted reports.

The system's web-based interface allows users to upload MRI scans, view segmentation and classification results, and generate detailed diagnostic reports. MediVision successfully bridges the gap between artificial intelligence and healthcare by providing a transparent, efficient, and accessible diagnostic support tool

FUTURE ENHANCEMENTS

Although MediVision successfully achieves accurate brain tumor segmentation and classification, several future improvements can be incorporated to enhance its performance, usability, and clinical integration. These enhancements will help extend the system beyond diagnosis and move toward intelligent healthcare assistance.

(i) Real-Time Analysis

The current model operates in batch mode for uploaded MRI scans. In the future, MediVision can be optimized for real-time tumor analysis, where results are generated instantly as MRI images are captured. This can be achieved through faster model inference, GPU optimization, and cloud deployment for on-demand processing.

(ii) Support for Multiple Imaging Modalities

Future upgrades can enable MediVision to handle multiple medical imaging types such as CT, PET, and fMRI scans. By combining features from different modalities, the system can produce more accurate and comprehensive diagnostic results, improving the reliability of tumor detection across various imaging environments.

(iii) Intelligent Report Generation

An enhanced version of MediVision can include an automated reporting module powered by Natural Language Processing (NLP). This would generate personalized, doctor-friendly summaries that clearly describe tumor type, affected region, and severity level, eliminating the need for manual interpretation of results.

(iv) Hospital System Integration

To improve real-world usability, MediVision can be integrated with hospital Electronic Health Record (EHR) systems. Such connectivity will allow doctors to retrieve patient history, update diagnostic results, and maintain long-term records directly from the web interface, ensuring efficient workflow and accurate patient tracking.

(v) Explainable and Transparent AI

The inclusion of Explainable AI (XAI) techniques such as Grad-CAM heatmaps or attention overlays can help visualize how the model interprets MRI images. This feature will increase the transparency of results and help radiologists verify the accuracy of AI-generated predictions, making the system more trustworthy in clinical use.

(vi) Advanced Clinical Assistance

Beyond detection and classification, *MediVision* can be expanded to assist in treatment planning. By tracking tumor growth, estimating volume changes, and correlating with patient recovery data, the system can offer insights that support personalized therapy recommendations and follow-up evaluations.

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